

Session 1 · Deck 1.A2

The Macro Research Frontier

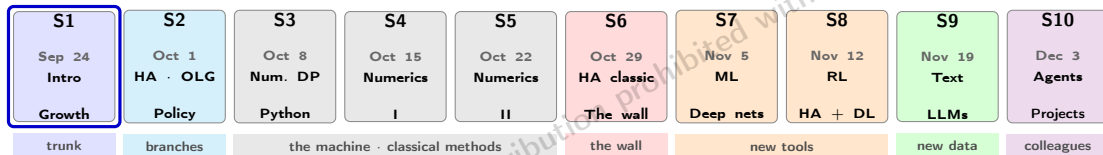
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Thursday, September 24, 2026

Build it on paper → solve it on the machine → push it to the frontier with AI

Where We Are

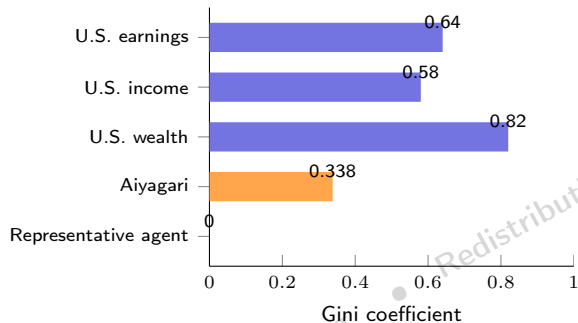
▼ you are here: 1.A2



Build it on paper → solve it on the machine → push it to the frontier with AI

This deck: what modern macroeconomics is built from, the questions each workhorse model answers, and why every step toward realism makes computation harder.

A Puzzle to Start: Why Is Wealth So Much More Unequal Than Income?



U.S.: SCF 2007, Díaz-Giménez, Glover & Ríos-Rull (2011). Model: the Stage-1A classical Aiyagari reference, Gini 0.338 — we replicate it in S6.

- The top 1% hold roughly **one third** of U.S. wealth.
- A representative agent has **no distribution at all**.
- The textbook incomplete-markets model gets **well under half** of the wealth Gini.

What must a model contain to get this right?
Persistent earnings risk? Entrepreneurs?
Health shocks? Bequests? Rate-of-return heterogeneity?

Every candidate answer is a **new state variable** — and a harder computation.

What Is Quantitative Macroeconomics?

- Macroeconomics that gives **numbers**, not only signs: how much, for whom, over what horizon.
- Three ingredients, always together:
 1. *Theory* — models with optimizing agents and equilibrium;
 2. *Data* — calibration and estimation;
 3. *Computation* — numerical solutions we can trust.
- This course adds a fourth: **AI**, as a solver, a measurement tool, and a research collaborator.



The Trunk: The Neoclassical Growth Model

Sequence problem: choose an infinite plan

$$\max_{\{c_t, k_{t+1}\}_{t \geq 0}} \sum_{t=0}^{\infty} \beta^t u(c_t)$$

s.t. $c_t + k_{t+1} = f(k_t) + (1 - \delta)k_t$, k_0 given.

Recursive problem: find one function

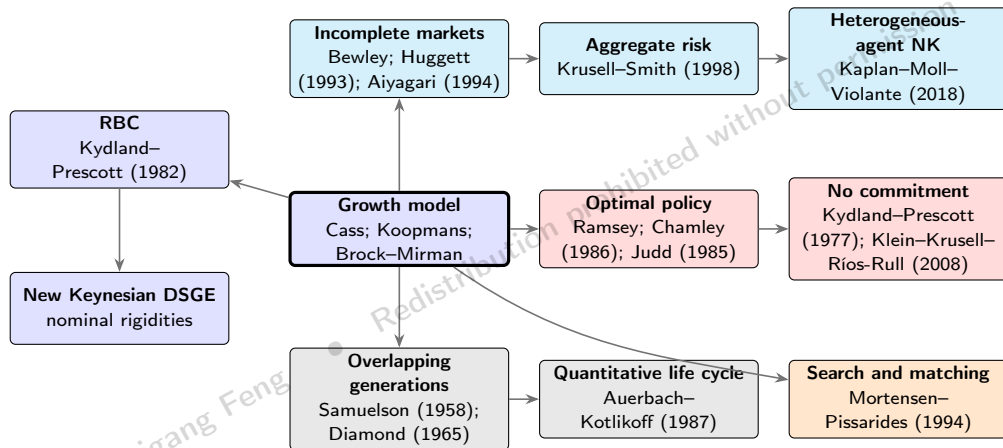
$$v(k) = \max_{k'} \left\{ u(f(k) + (1 - \delta)k - k') + \beta v(k') \right\}$$

solution: value $v(k)$ and policy $k' = g(k)$.



Almost every model in this course is this problem with **more states, more agents, more frictions, or a strategic government** added. Block B builds it carefully.

A Family Tree of Workhorse Models



Color key: representative agent heterogeneity labor frictions policy life cycle

None of these branches stopped in the year on its label — the next slide says where each one is now.

Where Each Branch Is Now

Branch	Where it has moved	Where to look
Incomplete markets	continuous time: an HJB equation for the household, a Kolmogorov forward equation for the distribution	Achdou, Han, Lasry, Lions & Moll (<i>REStud</i> 2022)
Aggregate risk	perturb <i>in the distribution</i> , or solve the whole path in sequence space	Reiter (2009); Auclert, Bardóczy, Rognlie & Straub (<i>ECMA</i> 2021)
HANK	two assets, liquid and illiquid, so measured MPCs can be matched	Kaplan, Moll & Violante (2018)
The wealth tail	entrepreneurs, bequests, heterogeneous returns	Quadrini (2000); Cagetti & De Nardi (2006); Benhabib, Bisin & Zhu (2011)
Optimal policy	the distribution inside the government's problem, without commitment	Feng, Han, Sargent & Zhu (2026), work in progress

And every branch now has a deep-learning frontier — Maliar, Maliar & Winant (2021); deep equilibrium nets (Azinovic, Gaegauf & Scheidegger 2022); DeepHAM (Han, Yang & E, *QE* 2026); DeepSAM (Payne, Rebei & Yang 2024). The rows above carry a distribution more cleverly; this line changes *what is carried* — functions **learned from sampled states** rather than tabulated on a grid.

Each Branch Answers a Question

Question	Model	What we solve for
Why are safe rates low when income risk is uninsurable?	Huggett (1993)	bond price clearing a credit market
Does precautionary saving raise capital? How unequal is wealth?	Aiyagari (1994)	r^* , K^* , stationary wealth distribution
How do inequality and business cycles interact?	Krusell–Smith (1998)	law of motion of the distribution
Who matches with whom, and how do jobs respond to shocks?	Search and matching	wages, match surplus, the matching distribution
Aging, social security, public debt burdens across cohorts?	Life-cycle OLG	cohort distributions, transition paths
What should a government tax — with or without commitment?	Ramsey / Markov-perfect	tax rules as functions of the state
Monetary policy when MPCs differ across households?	HANK	intertemporal MPCs, impulse responses

A good macro model is chosen by the question, not the other way round.

Heterogeneity Breaks the Simplicity

Representative agent

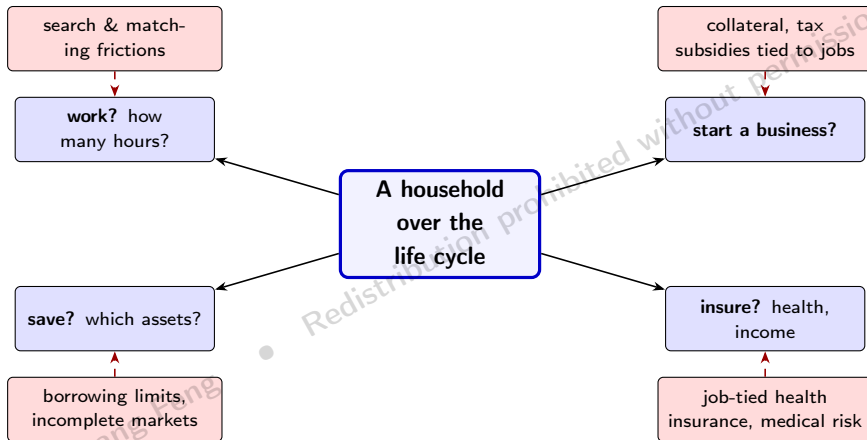
- Individual k equals aggregate K .
- Welfare theorems: solve a *planner's* problem instead of finding prices.
- State: (k, z) — two numbers.

Heterogeneous agents

- Households differ in assets a and income z ; prices depend on the **distribution** Γ : $r(\Gamma), w(\Gamma)$.
- Γ moves with everyone's choices: $\Gamma' = H(\Gamma; g)$.
- To forecast prices, agents must forecast Γ — an **infinite-dimensional** object.
- Welfare theorems fail: prices must be *solved for*.

Heterogeneity is not a detail you add at the end: it changes **what an equilibrium is** and **what we must compute**.

Frictions Shape Individual Choices — and the Aggregates

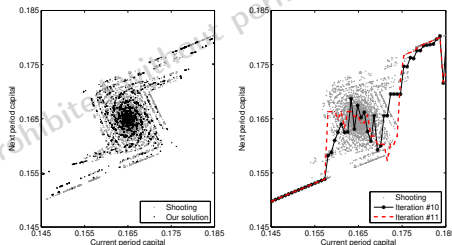


Each friction is a **constraint** or a **new state**: a job-match status, a health state, a business, an asset portfolio.

Example: job-tied health insurance distorts who works and who becomes an entrepreneur — visible only in a model with heterogeneity (1.A1, pillar iii).

The Government Is a Player Too

- **With commitment (Ramsey).** Choose the whole tax path at date 0. Private Euler equations become constraints, and past promises (a co-state Φ_{t-1}) become **extra state variables**.
- **Without commitment.** Each government re-optimizes; policy is a *rule* $\tau(\text{state})$ that must be a best response to its own future self — a fixed point in functions.
- **Theory warns us.** In a simple growth model with a state-dependent capital tax, a *continuous* Markov equilibrium can fail to exist; equilibrium can cycle (Santos, 2002).



The Santos (2002) example computed numerically: equilibrium paths (k_t vs. k_{t+1}) do not settle on a smooth policy function.

What Are We Solving For? The Objects Escalate

Model	Equilibrium object	Computational problem
Growth model	value $v(k)$ and policy $g(k)$	one fixed point (Bellman)
Huggett / Aiyagari	policies $g(a, z)$, a price r , a stationary distribution $\lambda(a, z)$	nested fixed point: household → distribution → price
Krusell–Smith	policies $g(a, z; \Gamma, Z)$ and a law of motion $\Gamma' = H(\Gamma, Z, Z')$	fixed point in <i>forecasting rules</i>
Life-cycle OLG	policies by age $g_j(a, z)$, cohort distributions, transition paths	backward (households) / forward (distributions)
Ramsey / Markov-perfect	tax rules $\tau(\text{state})$ consistent with private responses	fixed point in <i>policy rules</i> , with strategic interaction

Moving down the table, the object we compute goes from **a few numbers** to **functions of functions**. That is the arc of S2 (theory) and S3–S6 (computation).

Computation Is the Bottleneck

The curse of dimensionality (Bellman, 1957)

state dimension d	grid points (100^d)
1	10^2
3	10^6
6	10^{12}
10	10^{20}
distribution Γ	∞

- Each friction from two slides ago adds a dimension.
- A heterogeneous-agent economy with aggregate risk has a **distribution** in its state.
- An optimal-policy problem puts **all of that** inside the government's problem — and asks for a fixed point.

Grids and moments can only take us so far. Where they fail (S6), **sampling plus flexible function approximation** — neural networks and reinforcement learning — takes over (S7–S8).

...and a Numerical Answer Can Be Wrong in Ways That Look Right

- **Existence.** With frictions or distortions, a recursive equilibrium on the “natural” state space may not exist; one may need a larger state space.
- **Uniqueness.** Multiple stationary equilibria and multiple policy fixed points are common.
- **Continuity.** Policy functions can jump — and a smooth approximation will quietly paper over the jump. Where they do, an equilibrium can sometimes still be *constructed*, as the limit of approximate ones (Feng & Santos, “Markov Equilibria with Quasi-Hyperbolic Discounting”, working paper).
- **Accuracy.** A solver can converge to something that is *not* an equilibrium. Simulations of non-optimal economies need methods with convergence guarantees (Feng, Miao, Peralta-Alva & Santos, *IER* 2014).

Course rule: every computed result in this course comes with a **benchmark** (a closed form, a classical solution, or a published number) and an **accuracy check** (e.g. Euler-equation errors). No exceptions — AI-generated code included.

Takeaways

1. Quantitative macro gives **numbers you can defend**: theory + data + computation.
2. Nearly every workhorse model is the **growth model plus** heterogeneity, finite lives, frictions, or a strategic government.
3. Each extension changes **what an equilibrium is**: from numbers, to functions, to functions of distributions, to fixed points in policy rules.
4. Realism is paid for in **dimensions** — computation is the binding constraint on the questions we can ask.
5. Numerical answers can be wrong in ways that look right: **benchmark and verify**, always.

Bridge: Where Does AI Enter?

So far AI has barely appeared. It enters this story twice:

As a tool

Neural networks, reinforcement learning, language models and agents help us *solve* models past the wall, *measure* the economy, and *run* research.

As an object

AI is a new kind of capital: it reallocates tasks, reshapes wages and the tax base, and demands new models of growth and distribution.

Next (1.A3): AI as tool *and* object of macro research — and how this course works.